National
Qualifications
2023

2023 Mathematics

Higher – Paper 1

Finalised Marking Instructions

© Scottish Qualifications Authority 2023

These marking instructions have been prepared by examination teams for use by SQA appointed markers when marking external course assessments.

The information in this document may be reproduced in support of SQA qualifications only on a non-commercial basis. If it is reproduced, SQA must be clearly acknowledged as the source. If it is to be reproduced for any other purpose, written permission must be obtained from permissions@sqa.org.uk.

General marking principles for Higher Mathematics

Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.

For each question, the marking instructions are generally in two sections:

- *generic scheme — this indicates why each mark is awarded*
- *illustrative scheme — this covers methods which are commonly seen throughout the marking*

In general, you should use the illustrative scheme. Only use the generic scheme where a candidate has used a method not covered in the illustrative scheme.

(a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.

(b) If you are uncertain how to assess a specific candidate response because it is not covered by the general marking principles or the detailed marking instructions, you must seek guidance from your team leader.

(c) One mark is available for each •. There are no half marks.

(d) If a candidate's response contains an error, all working subsequent to this error must still be marked. Only award marks if the level of difficulty in their working is similar to the level of difficulty in the illustrative scheme.

(e) Only award full marks where the solution contains appropriate working. A correct answer with no working receives no mark, unless specifically mentioned in the marking instructions.

(f) Candidates may use any mathematically correct method to answer questions, except in cases where a particular method is specified or excluded.

(g) If an error is trivial, casual or insignificant, for example $6\times6=12$, candidates lose the opportunity to gain a mark, except for instances such as the second example in point (h) below.

(h) If a candidate makes a transcription error (question paper to script or within script), they lose the opportunity to gain the next process mark, for example

This is a transcription error and so the mark is not awarded.

This is no longer a solution of a quadratic equation, so the mark is not awarded.

$$x^2+5x+7=9x+4$$
$$x-4x+3=0$$
$$x=1$$

The following example is an exception to the above

This error is not treated as a transcription error, as the candidate deals with the intended quadratic equation. The candidate has been given the benefit of the doubt and all marks awarded.

$$x^2+5x+7=9x+4$$
$$x-4x+3=0$$
$$(x-3)(x-1)=0$$
$$x=1 \text{ or } 3$$

(i) Horizontal/vertical marking

If a question results in two pairs of solutions, apply the following technique, but only if indicated in the detailed marking instructions for the question.

Example:

	$\bullet^5$	$\bullet^6$
$\bullet^5$	$x=2$	$x=-4$
$\bullet^6$	$y=5$	$y=-7$

Horizontal: $\bullet^5$ $x=2$ and $x=-4$ $\quad\bullet^6$ $y=5$ and $y=-7$

Vertical: $\bullet^5$ $x=2$ and $y=5$ $\quad\bullet^6$ $x=-4$ and $y=-7$

You must choose whichever method benefits the candidate, **not** a combination of both.

(j) In final answers, candidates should simplify numerical values as far as possible unless specifically mentioned in the detailed marking instruction. For example

$\frac{15}{12}$ must be simplified to $\frac{5}{4}$. or $1\frac{1}{4}$ $\qquad$ $\frac{43}{1}$ must be simplified to 43

$\frac{15}{0.3}$ must be simplified to 50 $\qquad$ $\frac{4\not{5}}{3}$ must be simplified to $\frac{4}{15}$

$\sqrt{64}$ must be simplified to 8*

*The square root of perfect squares up to and including 144 must be known.

(k) Commonly Observed Responses (COR) are shown in the marking instructions to help mark common and/or non-routine solutions. CORs may also be used as a guide when marking similar non-routine candidate responses.

(l) Do not penalise candidates for any of the following, unless specifically mentioned in the detailed marking instructions:

- working subsequent to a correct answer
- correct working in the wrong part of a question
- legitimate variations in numerical answers/algebraic expressions, for example angles in degrees rounded to nearest degree
- omission of units
- bad form (bad form only becomes bad form if subsequent working is correct), for example
  $\left(x^3+2x^2+3x+2\right)(2x+1)$ written as
  $\left(x^3+2x^2+3x+2\right)\times 2x+1$
  $=2x^4+5x^3+8x^2+7x+2$ gains full credit
- repeated error within a question, but not between questions or papers

(m) In any 'Show that…' question, where candidates have to arrive at a required result, the last mark is not awarded as a follow-through from a previous error, unless specified in the detailed marking instructions.

(n) You must check all working carefully, even where a fundamental misunderstanding is apparent early in a candidate's response. You may still be able to award marks later in the question so you must refer continually to the marking instructions. The appearance of the correct answer does not necessarily indicate that you can award all the available marks to a candidate.

(o) You should mark legible scored-out working that has not been replaced. However, if the scored-out working has been replaced, you must only mark the replacement working.

(p) If candidates make multiple attempts using the same strategy and do not identify their final answer, mark all attempts and award the lowest mark. If candidates try different valid strategies, apply the above rule to attempts within each strategy and then award the highest mark.

For example:

Strategy 1 attempt 1 is worth 3 marks.	Strategy 2 attempt 1 is worth 1 mark.
Strategy 1 attempt 2 is worth 4 marks.	Strategy 2 attempt 2 is worth 5 marks.
From the attempts using strategy 1, the resultant mark would be 3.	From the attempts using strategy 2, the resultant mark would be 1.

In this case, award 3 marks.

Marking instructions for each question

Question	Generic scheme	Illustrative scheme	Max mark
1.	$\bullet^1$ express second term in differentiable form	$\bullet^1$ $\ldots -10x^{-4}$ stated or implied by $\bullet^3$	3
	$\bullet^2$ differentiate one term	$\bullet^2$ $\frac{5}{3}x^{\frac{2}{3}}\ldots$ or $\ldots +40x^{-5}$	
	$\bullet^3$ complete differentiation	$\bullet^3$ $\frac{5}{3}x^{\frac{2}{3}}+40x^{-5}$	

Notes:

1. Where candidates "differentiate over two lines" see Candidates A and B.
2. $\bullet^3$ is only available for differentiating a term with a negative index.
3. Where candidates attempt to integrate throughout, only $\bullet^1$ is available.

Commonly Observed Responses:

Candidate A - differentiating over two lines

$y = x^{\frac{5}{3}} - \frac{10}{x^4}$

$y = \frac{5}{3}x^{\frac{2}{3}} - 10x^{-4}$ $\bullet^1$ ✓

$y = \frac{5}{3}x^{\frac{2}{3}} + 40x^{-5}$ $\bullet^2$ ✓ $\bullet^3$ ✗

Candidate B - differentiating over two lines

$y = x^{\frac{5}{3}} - \frac{10}{x^4}$

$y = \frac{5}{3}x^{\frac{2}{3}} - 10x^{-4}$ $\bullet^1$ ✓

$y = \frac{5}{3}x^{\frac{2}{3}} + 40x^{-3}$ $\bullet^2$ ✓ $\bullet^3$ ✗

Candidate C

$\vdots$

$\frac{5}{3}x^{\frac{2}{3}} + 40x^{-5} + c$ $\bullet^3$ ✗

Question	Generic scheme	Illustrative scheme	Max mark
2.	$\bullet^1$ find midpoint of PQ	$\bullet^1$ $(4,3)$	**4**
	$\bullet^2$ calculate gradient of PQ	$\bullet^2$ $-\frac{1}{2}$ or $-\frac{6}{12}$	
	$\bullet^3$ state perpendicular gradient	$\bullet^3$ 2 stated or implied by $\bullet^4$	
	$\bullet^4$ determine equation of perpendicular bisector	$\bullet^4$ $y=2x-5$	

Notes:

1. $\bullet^4$ is only available as a consequence of using a perpendicular gradient **and** a midpoint.
2. The gradient of the perpendicular bisector must appear in fully simplified form at $\bullet^3$ or $\bullet^4$ stage for $\bullet^4$ to be awarded.
3. At $\bullet^4$, accept $2x-y=5$, $y-2x=-5$ or any other rearrangement of the equation where the constant terms have been simplified.

Commonly Observed Responses:

Question	Generic scheme	Illustrative scheme	Max mark
3.	**Method 1** •1 apply $\log_5 x - \log_5 y = \log_5 \frac{x}{y}$ •2 write in exponential form •3 process for x	**Method 1** •1 $\log_5 \frac{x}{3} \ldots$ •2 $\frac{x}{3} = 5^2$ •3 75	3
	Method 2 •1 apply $\log_5 x - \log_5 y = \log_5 \frac{x}{y}$ •2 apply $m \log_5 x = \log_5 x^m$ •3 process for x	**Method 2** •1 $\log_5 \frac{x}{3} \ldots$ •2 $\ldots = \log_5 5^2$ •3 75	3

Notes:

1. Each line of working must be equivalent to the line above within a valid strategy, however see Candidates A and B for exceptions.
2. Where candidates do not use exponentials at •2 , •3 is not available – see Candidate C.

Commonly Observed Responses:

Candidate A - incorrect exponential

$\log_5 \frac{x}{3} = 2$	•1 ✓
$\frac{x}{3} = 2^5$	•2 ✗
$x = 96$	•3 ✓1

Candidate B

$\log_5 3x = 2$	•1 ✗
$3x = 5^2$	•2 ✓1
$x = \frac{25}{3}$	•3 ✓1

Candidate C - no use of exponentials

$\log_5 \frac{x}{3} = 2$	•1 ✓
$\frac{x}{3} = 10$	•2 ✗
$x = 30$	•3 ✗

Question			Generic scheme	Illustrative scheme	Max mark
4.	(a)		$\bullet^1$ find $\cos p$	$\bullet^1$ $\frac{3}{5}$	**1**
			$\bullet^2$ find $\cos q$	$\bullet^2$ $\frac{3}{\sqrt{45}}\left(=\frac{1}{\sqrt{5}}\right)$	**1**
Notes:					
1. Accept $\frac{3}{3\sqrt{5}}$ for $\bullet^2$.					
Commonly Observed Responses:					
	(b)		$\bullet^3$ select appropriate formula and express in terms of p and q $\bullet^4$ substitute into addition formula $\bullet^5$ evaluate $\cos(p+q)$	$\bullet^3$ $\cos p\cos q-\sin p\sin q$ $\bullet^4$ $\frac{3}{5}\times\frac{3}{\sqrt{45}}-\frac{4}{5}\times\frac{6}{\sqrt{45}}$ $\bullet^5$ $-\frac{3}{\sqrt{45}}\left(=-\frac{1}{\sqrt{5}}\right)$	**3**
Notes:					
2. Award $\bullet^3$ for candidates who write $\cos\left(\frac{3}{5}\right)\times\cos\left(\frac{3}{\sqrt{45}}\right)-\sin\left(\frac{4}{5}\right)\times\sin\left(\frac{6}{\sqrt{45}}\right)$. $\bullet^4$ and $\bullet^5$ are unavailable. 3. For any attempt to use $\cos(p+q)=\cos p\pm\cos q$, $\bullet^4$ and $\bullet^5$ are unavailable. 4. $\bullet^5$ is only available if either the surd part or the non-surd part of the fraction is simplified as far as possible. Accept $-\frac{3}{\sqrt{45}}$, $-\frac{\sqrt{45}}{15}$, $-\frac{15}{15\sqrt{5}}$ or answers obtained on follow through which do not require simplification. Do not accept $-\frac{15}{5\sqrt{45}}$. 5. $\bullet^5$ is only available for an answer expressed as a single fraction.					
Commonly Observed Responses:					

Question	Generic scheme	Illustrative scheme	Max mark
5.	•1 use the discriminant •2 apply condition and express in standard quadratic form •3 process for p	•1 $(3p-2)^2-4\times2\times p$ •2 $9p^2-20p+4=0$ •3 $\frac{2}{9}, 2$	**3**

Notes:

1. Where candidates states an incorrect condition, •2 is not available. However, •3 is available for finding the roots of the quadratic - see Candidate B.
2. Where x appears in any expression, no further marks are available.

Commonly Observed Responses:

Candidate A

(For equal roots) $b^2-4ac=0$

$(3p-2)^2-4\times2\times p$ •1 ✓

$9p^2-20p+4$ •2 ✓

$p=\frac{2}{9}, 2$ •3 ✓

Candidate B

(For equal roots) $b^2-4ac>0$ •2 ✗

$(3p-2)^2-4\times2\times p$ •1 ✓

$9p^2-20p+4=0$

$p=\frac{2}{9}, 2$ •3 ✓1

⋮

Question	Generic scheme	Illustrative scheme	Max mark
6.	$\bullet^1$ express second term in integrable form	$\bullet^1$ $\ldots - 6x^{\frac{1}{2}}$	4
	$\bullet^2$ integrate one term	$\bullet^2$ $\frac{2}{6}x^6 \ldots$ or $\ldots - \frac{6x^{\frac{3}{2}}}{\frac{3}{2}}$	
	$\bullet^3$ integrate other term	$\bullet^3$ $\ldots - \frac{6x^{\frac{3}{2}}}{\frac{3}{2}}$ or $\frac{2}{6}x^6 \ldots$	
	$\bullet^4$ complete integration	$\bullet^4$ $\frac{1}{3}x^6 - 4x^{\frac{3}{2}} + c$	

Notes:

1. The mark for integrating the final term is only available if candidates integrate a term with a fractional index.
2. Do not penalise the appearance of an integral sign and/or dx throughout.
3. Do not penalise the omission of '$+c$' at $\bullet^2$ or $\bullet^3$.
4. All coefficients must be simplified at $\bullet^4$ stage for $\bullet^4$ to be awarded.
5. Accept $\frac{x^6 - 12x^{\frac{3}{2}}}{3} + c$ for $\bullet^4$ but do not accept $\frac{2x^6 - 24x^{\frac{3}{2}}}{6} + c$.
6. $\bullet^2$, $\bullet^3$ and $\bullet^4$ are not available within an invalid strategy.

Commonly Observed Responses:

Candidate A

$\int \left(2x^5 - 6x^{\frac{1}{2}}\right) dx$ $\bullet^1$ ✓

$= \frac{2x^6}{6} - \frac{6x^{\frac{3}{2}}}{\frac{3}{2}} + c$ $\bullet^2$ ✓ $\bullet^3$ ✓

$= \frac{2x^6}{6} - 4x^{\frac{3}{2}} + c$

$= \frac{1}{3}x^6 - 4\sqrt{x} + c$ $\bullet^4$ ×

$\bullet^4$ cannot be awarded over two lines of working

Candidate B - integrating over two lines

$\frac{2x^6}{6} - 6x^{\frac{1}{2}}$ $\bullet^1$ ✓

$= \frac{2x^6}{6} - \frac{6x^{\frac{3}{2}}}{\frac{3}{2}} + c$ $\bullet^2$ ✓ $\bullet^3$ ×

$= \frac{1}{3}x^6 - 4x^{\frac{3}{2}} + c$ $\bullet^4$ ✓1

Candidate C - insufficient evidence

$\int 2x^5 - 6x^{\frac{1}{2}} dx$ $\bullet^1$ ✓

$\frac{1}{3}x^6 - 9x^{\frac{3}{2}} + c$ $\bullet^2$ ✓ $\bullet^3$ ^ $\bullet^4$ ×

Candidate D

⋮ $\bullet^1$ ✓

$= \frac{1}{3}x^6 - 4x^{\frac{3}{2}}$ $\bullet^2$ ✓ $\bullet^3$ ✓

$= \frac{1}{3}x^6 - 4\sqrt{x^3} + c$ $\bullet^4$ ✓

Question			Generic scheme	Illustrative scheme	Max mark
7.	(a)		•[1] use laws of logs •[2] evaluate log	•[1] $\log_2 \frac{5}{40}$ •[2] -3	**2**
Notes:					
1. Do not penalise the omission of the base of the logarithm at •[1]. 2. Correct answer with no working, award 0/2.					
Commonly Observed Responses:					
Candidate A - introducing a variable $\log_2\left(5\times\frac{1}{40}\right)$ •[1] ✓ $\log_2 \frac{1}{8}$ $2^x = \frac{1}{8}$ $x=-3$ •[2] ✓					
	(b)		•[3] state range	•[3] $0<a<1$	**1**
Notes:					
3. At •[3] accept "$a>0$ and $a<1$" or "$a>0$, $a<1$ ".					
Commonly Observed Responses:					

Question	Generic scheme	Illustrative scheme	Max mark
8.	$\bullet^1$ start to differentiate	$\bullet^1$ $3x^2\ldots$ or $\ldots+6x\ldots$ or $\ldots-9$	**6**
	$\bullet^2$ complete differentiation and equate to 0	$\bullet^2$ $3x^2+6x-9=0$	
	$\bullet^3$ solve for x	$\bullet^3$ -3 and 1	
	$\bullet^4$ process for y	$\bullet^4$ 32 and 0	
	$\bullet^5$ construct nature table(s)	$\bullet^5$ $\bullet^5$ $\bullet^6$ x | $\ldots$ | -3 | $\ldots$ | 1 | $\ldots$ $f'(x)$ | $+$ | 0 | $-$ | 0 | $+$ shape | / | — | \ | — | /	
	$\bullet^6$ interpret and state conclusions	$\bullet^6$ max at $(-3,32)$; min at $(1,0)$	

Notes:

1. For a numerical approach award 0/6.
2. $\bullet^2$ is only available if '$=0$' appears at the $\bullet^2$ stage or in working leading to $\bullet^3$, however see Candidates A and B.
3. Candidates who equate their derivative to 0, may use division by 3 as a strategy - see candidates B, C and D.
4. $\bullet^3$ is available to candidates who factorise **their** derivative from $\bullet^2$ as long as it is of equivalent difficulty.
5. $\bullet^3$ and $\bullet^4$ may be awarded vertically.
6. $\bullet^5$ is not available where any errors are made in calculating values of $f'(x)$.
7. $\bullet^5$ and $\bullet^6$ may be awarded vertically.
8. $\bullet^6$ is still available in cases where a candidates table of signs does not lead legitimately to a maximum/minimum shape.
9. Candidates may use the second derivative - see Candidates E and F.
10. Accept "max when $x=-3$" **and** "min when $x=1$" for $\bullet^6$.

Commonly Observed Responses:

Candidate A

Stationary points when $f'(x)=0$

$f'(x)=3x^2+6x-9$ $\bullet^1$ ✓ $\bullet^2$ ✓

$f'(x)=3(x+3)(x-1)$

$x=-3,\ 1$ $\bullet^3$ ✓

Candidate B

Stationary points when $f'(x)=0$

$f'(x)=3x^2+6x-9$ $\bullet^1$ ✓ $\bullet^2$ ✓

⋮

$f'(x)=(x+3)(x-1)$

$x=-3,\ 1$ $\bullet^3$ ✓

Candidate C - division by 3

$3x^2+6x-9=0$ $\bullet^1$ ✓ $\bullet^2$ ✓

$x^2+2x-3=0$

$x=-3,\ 1$ $\bullet^3$ ✓

Candidate D - derivative never equated to 0

$3x^2+6x-9$ $\bullet^1$ ✓ $\bullet^2$ ^

$x^2+2x-3=0$

$x=-3,\ 1$ $\bullet^3$ ✓1

Question	Generic scheme	Illustrative scheme	Max mark

8.(continued)

Commonly Observed Responses:

Candidate E - second derivative

$f''(x)=6x+6$

$\bullet^5$ ✓ $\bullet^6$ ✓

$f''(-3)<0$ $f''(1)>0$ $\bullet^5$ ✓

so max at $(-3,32)$ so min at $(1,0)$ $\bullet^6$ ✓

Candidate F - second derivative

$f''(x)=6x+6$

$\bullet^5$ ✓ $\bullet^6$ ✓

$f''(-3)=-12$, $f''(1)=12$ $\bullet^5$ ✓

$-12<0$ $12>0$

so max at $(-3,32)$ so min at $(1,0)$ $\bullet^6$ ✓

For the table of signs for a derivative, accept:

x	-3^-	-3	-3^+
$f'(x)$	+	0	−
Slope or shape	↗	—	↘

AND

x	1^-	1	1^+
$f'(x)$	−	0	+
Slope or shape	↘	—	↗

x	$\rightarrow$	-3	$\rightarrow$
$f'(x)$	+	0	−
Slope or shape	↗	—	↘

Arrows are taken to mean 'in the neighbourhood of'

AND

x	$\rightarrow$	1	$\rightarrow$
$f'(x)$	−	0	+
Slope or shape	↘	—	↗

Arrows are taken to mean 'in the neighbourhood of'

x	a	-3	b
$f'(x)$	+	0	−
Slope or shape	↗	—	↘

Where $a<-3$ and $-3<b<1$

AND

x	c	1	d
$f'(x)$	−	0	+
Slope or shape	↘	—	↗

Where $-3<c<1$ and $d>1$

For the table of signs for a derivative, accept:

x	$\rightarrow$	-3	$\rightarrow$	1	$\rightarrow$
$f'(x)$	+	0	−	0	+
Slope or shape	↗	—	↘	—	↗

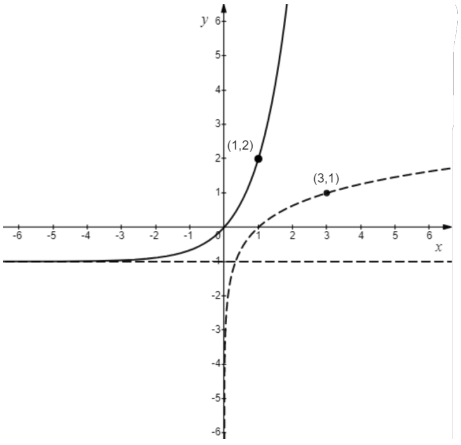
Since the function is continuous $-3\rightarrow 1$ is acceptable

x	a	-3	b	1	c
$f'(x)$	+	0	−	0	+
Slope or shape	↗	—	↘	—	↗

Since the function is continuous $-3<b<1$ is acceptable

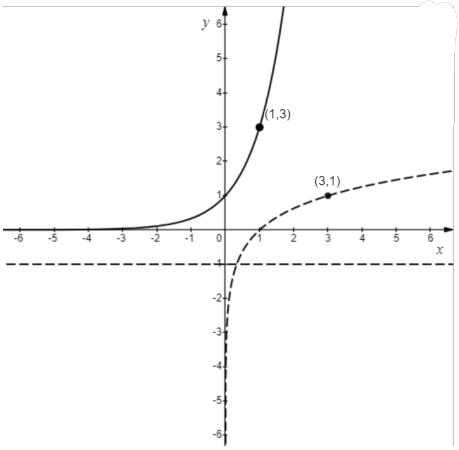
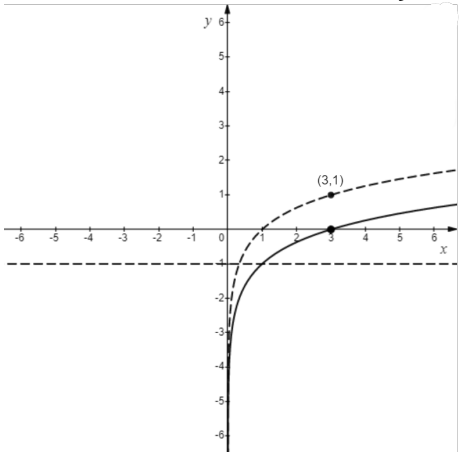
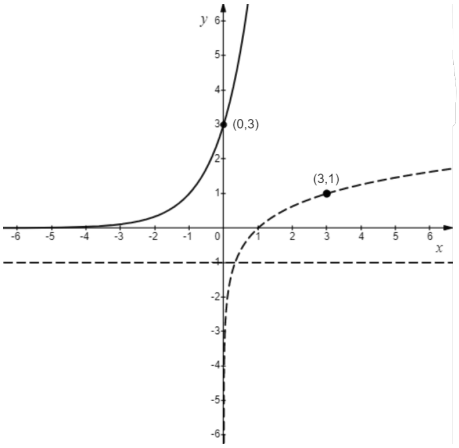
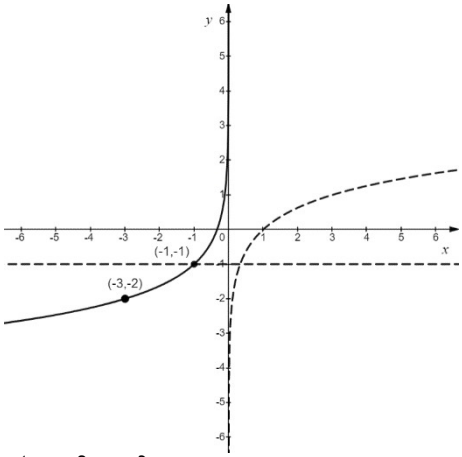
- For this question do not penalise the omission of 'x' or the word 'shape'/'slope'.
- Stating values of $f'(x)$ is an acceptable alternative to writing '+' or '−' signs.
- Acceptable variations of $f'(x)$ are: f', $\frac{df}{dx}$, $\frac{dy}{dx}$, $3x^2+6x-9$ and $3(x+3)(x-1)$

  but NOT x^2+2x-3 or $(x+3)(x-1)$.

Question			Generic scheme	Illustrative scheme	Max mark
9.			•[1] graph reflected in $y = x$ •[2] vertical translation of "-1" unit following a reflection in $y = x$ identifiable from graph •[3] sketch of required function	•[1] a concave up curve above the x-axis for $x > 0$ •[2] curve passing through (0,0) and (1,2) •[3] curve approaches the line $y = -1$ from above as $x \to -\infty$ 	**3**

Notes:

1. For •[1] accept any graph of a function which is concave up within the first quadrant.
2. •[1] is only available where the candidate has attempted to reflect the given curve in the line $y = x$.
3. •[3] is only available where the curve passes through (0,0) and (1,2).
4. The line $y = -1$ does not need to be shown.
5. For a rotation, award 0/3 – for example see Candidate D.

Question	Generic scheme	Illustrative scheme	Max mark
9.(continued)			
Commonly Observed Responses:			
Candidate A - reflection only  $\bullet^1$ ✓ $\bullet^2$ × $\bullet^3$ ×		**Candidate B - translation only**  $\bullet^1$ × $\bullet^2$ × $\bullet^3$ ×	
Candidate C - incorrect order of transformations  $\bullet^2$ × $\bullet^1$ ✓$_1$ $\bullet^3$ ×		Candidate D – rotation  $\bullet^1$ × $\bullet^2$ × $\bullet^3$ ×	

Question	Generic scheme	Illustrative scheme	Max mark
10. (a)	$\bullet^1$ use -5 in synthetic division or evaluation of quartic	$\bullet^1$ $-5 \mid 1 \quad 3 \quad -7 \quad 9 \quad -30$ $\quad\quad\quad 1 \quad\quad\quad\quad\quad \square$ or $(-5)^4+3\times(-5)^3-7\times(-5)^2$ $+9\times(-5)-30$	**2**
	$\bullet^2$ complete division/evaluation and interpret result	$\bullet^2$ $-5 \mid 1 \quad 3 \quad -7 \quad 9 \quad -30$ $\quad\quad\quad -5 \quad 10 \quad -15 \quad 30$ $\quad\quad\quad 1 \quad -2 \quad 3 \quad -6 \quad \boxed{0}$ Remainder $=0 \therefore (x+5)$ is a factor **OR** $f(-5)=0 \therefore (x+5)$ is a factor	

Notes:

1. Communication at $\bullet^2$ must be consistent with working at that stage i.e. a candidate's working must arrive legitimately at 0 before $\bullet^2$ can be awarded.
2. Accept any of the following for $\bullet^2$:
   - '$f(-5)=0$ so $(x+5)$ is a factor'
   - 'since remainder $=0$, it is a factor'
   - the '0' from any method linked to the word 'factor' by 'so', 'hence', $\therefore$, $\rightarrow$, $\Rightarrow$ etc.
3. Do not accept any of the following for $\bullet^2$:
   - double underlining the '0' or boxing the '0' without comment
   - '$x=-5$ is a factor', '... is a root'
   - the word 'factor' only, with no link.

Commonly Observed Responses:

Candidate A - grid method

	x^3			
x	x^4	$-2x^3$		
5	$5x^3$			

$\bullet^1$ ✓

	x^3	$-2x^2$	$+3x$	-6
x	x^4	$-2x^3$	$+3x^2$	$-6x$
5	$5x^3$	$-10x^2$	$+15x$	-30

with no remainder

$\therefore (x+5)$ is a factor $\bullet^2$ ✓

Candidate B - grid method

	x^3			
x	x^4	$-2x^3$		
5	$5x^3$			

$\bullet^1$ ✓

	x^3	$-2x^2$	$+3x$	-6
x	x^4	$-2x^3$	$+3x^2$	$-6x$
5	$5x^3$	$-10x^2$	$+15x$	-30

$\therefore (x+5)(x^3-2x^2+3x-6)=x^4+3x^3-7x^2+9x-30$

$\therefore (x+5)$ is a factor $\bullet^2$ ✓

Question			Generic scheme	Illustrative scheme	Max mark
10.	(b)		$\bullet^3$ identify cubic and attempt to factorise	$\bullet^3$ eg $1 \mid 1 \quad -2 \quad 3 \quad -6$ $\quad\quad\quad 1 \quad -1$ $\quad 1 \quad -1 \quad \square$ or $2 \mid 1 \quad -2 \quad 3 \quad -6$ $\quad\quad\quad 2 \quad 0$ $\quad 1 \quad 0 \quad \square$	**5**
			$\bullet^4$ find second factor	$\bullet^4$ $2 \mid 1 \quad -2 \quad 3 \quad -6$ $\quad\quad\quad 2 \quad 0 \quad 6$ $\quad 1 \quad 0 \quad 3 \mid 0$ leading to $(x-2)$ or $x=2$	
			$\bullet^5$ identify quadratic	$\bullet^5$ x^2+3	
			$\bullet^6$ interpret lack of solutions of quadratic	$\bullet^6$ $b^2-4ac=-12<0$ $\therefore$ no (further real) solutions OR $x^2=-3$ or ~~$x^2=-3$~~ $\therefore$ no (further real) solutions	
			$\bullet^7$ state solutions	$\bullet^7$ $x=-5,\ x=2$	

Notes:

4. Candidates who arrive at $(x+5)(x-2)(x^2+3)$ by using algebraic long division or by inspection gain $\bullet^3$, $\bullet^4$ and $\bullet^5$.
5. Evidence for $\bullet^6$ may appear in the quadratic formula.
6. At $\bullet^6$ accept interpretations such as "no further roots", "no solutions" and "cannot factorise further" with justification.
7. At $\bullet^6$ accept $x=\sqrt{-3}$ **leading to** "not possible" and "not real".
8. Where there is no reference to b^2-4ac accept '$-12<0$ so no real roots' **with the remaining roots stated** for $\bullet^6$ - see candidates E and F.
9. Do not accept any of the following for $\bullet^6$:
   - $(x+5)(x-2)(x^2+3)$ no further roots/cannot factorise further.
   - $(x+5)(x-2)(\ldots\ \ldots)(\ldots\ \ldots)$ no further roots/cannot factorise further.
10. Where the quadratic factor obtained at $\bullet^5$ can be factorised, $\bullet^6$ and $\bullet^7$ are not available.
11. $\bullet^7$ is only available where $\bullet^6$ has been awarded.

Looking for more resources? Visit scotpapers.com – Scotland's #1 Past Paper Archive

Question	Generic scheme	Illustrative scheme	Max mark

10.(continued)

Commonly Observed Responses:

Candidate C

$(x+5)(x-2)(x^2+3)$ •5 ✓

$b^2-4ac=0-12<0$ •6 ✓

so no solutions

$x=-5,\ x=2$ •7 ✓

Candidate D

$(x+5)(x-2)(x^2+3)$ •5 ✓

$b^2-4ac<0$ •6 ^

so no solutions

$x=-5,\ x=2$ •7 ✓2

Candidate E

$(x+5)(x-2)(x^2+3)$ •5 ✓

$-12<0$ •6 ✓

so no solutions

$x=-5,\ x=2$ •7 ✓

Candidate F

$(x+5)(x-2)(x^2+3)$ •5 ✓

$-12<0$

so no solutions •6 ^ •7 ^

Candidate G - grid method

(a)

	x^3	$-2x^2$	$3x$	-6
x	x^4	$-2x^3$	$+3x^2$	$-6x$
5	$5x^3$	$-10x^2$	$+15x$	-30

(b)

	x^2	…	…
x	x^3	…	…
…	…	…	…

•3 ✓

•3 is awarded for evidence of the cubic expression (which may be in the grid from part (a)) **AND** the terms in the diagonal boxes summing to the second and third terms in the cubic respectively.

	x^2	$0x$	3
x	x^3	0	$3x$
-2	$-2x^2$	0	-6

•4 ✓

$(x+5)(x-2)(x^2+3)$ •5 ✓

$b^2-4ac=-12<0$

$\therefore$ no more solutions •6 ✓

$x=-5,\ x=2$ •7 ✓

Question			Generic scheme	Illustrative scheme	Max mark
11.	(a)		$\bullet^1$ integrate $\bullet^2$ substitute limits $\bullet^3$ evaluate integral	$\bullet^1$ $-5\cos x - 3\sin x$ $\bullet^2$ $\left[-5\cos\pi - 3\sin\pi\right]$ $-\left[-5\cos\frac{\pi}{2} - 3\sin\frac{\pi}{2}\right]$ $\bullet^3$ 8	3

Notes:

1. Where candidates make no attempt to integrate or use another invalid approach award 0/3 - see Candidate A. However see also Candidates B to F.
2. Do not penalise the inclusion of '$+c$' or the continued appearance of the integral sign.
3. Candidates who change the limits to degrees before integrating cannot gain $\bullet^1$. However, $\bullet^2$ and $\bullet^3$ are still available.
4. $\bullet^3$ is only available where candidates have considered both limits within a trigonometric function.
5. The minimum acceptable response for $\bullet^2$ is $5-(-3)$.

Commonly Observed Responses:

Candidate A - introducing a power

Eg $5\sin x^2 - 3\cos x^2$ $\bullet^1$ ✗ $\bullet^2$ ✗ $\bullet^3$ ✗

⋮

Candidate B - differentiating in full

$5\cos x + 3\sin x$ $\bullet^1$ ✗

$\left(5\cos\pi + 3\sin\pi\right) - \left(5\cos\frac{\pi}{2} + 3\sin\frac{\pi}{2}\right)$ $\bullet^2$ ✓2

-8 $\bullet^3$ ✓1

Candidate C - integrating one term

$5\cos x - 3\sin x$ $\bullet^1$ ✗

$\left(5\cos\pi - 3\sin\pi\right) - \left(5\cos\frac{\pi}{2} - 3\sin\frac{\pi}{2}\right)$ $\bullet^2$ ✓1

-2 $\bullet^3$ ✓1

Candidate D - integrating one term

$-5\cos x + 3\sin x$ $\bullet^1$ ✗

$\left(-5\cos\pi + 3\sin\pi\right) - \left(-5\cos\frac{\pi}{2} + 3\sin\frac{\pi}{2}\right)$ $\bullet^2$ ✓1

2 $\bullet^3$ ✓1

Candidate E - integrating one term

Eg $5\sin x - 3\sin x$ $\bullet^1$ ✗

$\left(5\sin\pi - 3\sin\pi\right) - \left(5\sin\frac{\pi}{2} - 3\sin\frac{\pi}{2}\right)$ $\bullet^2$ ✓1

-2 $\bullet^3$ ✓2

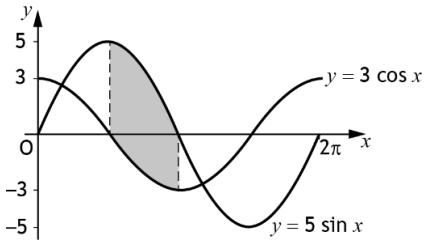
Mark 3 is not of equivalent difficulty - only 2 exact values

Candidate F - obtaining other expressions of the form $a\sin x + b\cos x$

Eg $-\frac{1}{5}\cos x - \frac{1}{3}\sin x$ $\bullet^1$ ✗

$\left(-\frac{1}{5}\cos\pi - \frac{1}{3}\sin\pi\right) - \left(-\frac{1}{5}\cos\frac{\pi}{2} - \frac{1}{3}\sin\frac{\pi}{2}\right)$ $\bullet^2$ ✓2

$\frac{8}{15}$ $\bullet^3$ ✓1

Question			Generic scheme	Illustrative scheme	Max mark
11.	(b)		$\bullet^4$ identify boundaries and shade area	$\bullet^4$ 	**1**
Notes:					
Commonly Observed Responses:					

Question	Generic scheme	Illustrative scheme	Max mark
12.	**Method 1** •1 identify common factor •2 complete the square •3 process for c and write in required form	**Method 1** •1 $-2\left(x^2+6x\ldots\right.$ stated or implied by •2 •2 $-2(x+3)^2\ldots$ •3 $-2(x+3)^2+25$	3
	Method 2 •1 expand completed square form •2 equate coefficients •3 process for b and c and write in required form	**Method 2** •1 $ax^2+2abx+ab^2+c$ stated or implied by •2 •2 $a=-2,\ 2ab=-12,$ and $ab^2+c=7$ •3 $-2(x+3)^2+25$	

Notes:

1. $-2(x+3)^2+25$ with no working gains •1 and •2 only. However, see Candidate E.
2. •1 and •3 are not available in cases where $a>0$. For example, see Candidate F.

Commonly Observed Responses:

Candidate A

$-2(x^2+6)+7$

$-2\left((x+3)^2-9\right)+7$ •1 ✓ •2 ✓

$-2(x+3)^2+25$ •3 ✓

See the exception to marking principle (h)

Candidate B

$ax^2+2abx+ab^2+c$ •1 ✓

$a=-2,\ 2ab=-12,\ ab^2+c=7$ •2 ✓

$b=3,\ c=25$ •3 ^

•3 is lost as answer is not in completed square form

Candidate C

$-2\left(x^2+12x\right)+7$ •1 ✗

$-2\left((x+6)^2-36\right)+7$ •2 ✓1

$-2(x+6)^2+79$ •3 ✓1

Candidate D

$-2\left((x+6)^2-36\right)+7$ •1 ✗ •2 ✗

$-2(x+6)^2+79$ •3 ✓1

Candidate E

$-2(x+3)^2+25$ •1 ✓ •2 ✓

Check: $=-2\left(x^2+6x+9\right)+25$

$=-2x^2-12x-18+25$

$=-2x^2-12x+7$ •3 ✓

Candidate F

$-2x^2-12x+7$

$=2x^2+12x-7$ •1 ✗

$=2\left(x^2+6x\ldots\right.$

$=2(x+3)^2\ldots$ •2 ✓1

$=-2(x+3)^2\ldots$ •3 ✗

Question			Generic scheme	Illustrative scheme	Max mark
13.	(a)	(i)	•1 state exact value	•1 $\sqrt{3}$	1
		(ii)	•2 interpret notation •3 state expression for $f\left(g\left(x\right)\right)$	•2 $f\left(2x\right)$ or $2\sin\left(g\left(x\right)\right)$ •3 $2\sin 2x$	2

Notes:

1. For $f\left(g\left(x\right)\right)=2\sin 2x$ without working, award both •2 and •3.
2. Working for (a)(ii) may be found in (a)(i)

Commonly Observed Responses:

Candidate A

(a)(ii) $f\left(g\left(x\right)\right)=4\sin x$ •2 ✗ •3 ✓1

Candidate B - Beware of "2 attempts"

$f\left(g\left(x\right)\right)=2\sin x$ •2 ✗ •3 ✗

$f\left(2x\right)=2\sin 2x$

	(b)	(i)	•4 find the value of $\sin p$	•4 $\frac{1}{6}$	1
		(ii)	•5 expand $f\left(g\left(p\right)\right)$ using double angle formula •6 find value of $\cos p$ •7 substitute and determine exact value	•5 $2\times 2\sin p\cos p$ or $4\sin p\cos p$ **stated explicitly** •6 $\frac{\sqrt{35}}{6}$ •7 $2\times 2\times\frac{1}{6}\times\frac{\sqrt{35}}{6}$ leading to $\frac{\sqrt{35}}{9}$	3

Notes:

1. •5 is not available for expansions which do not involve p. •6 and •7 are still available.
   However, accept $\sin^{-1}\left(\frac{1}{6}\right)$ in place of p - see Candidate C.
2. •7 is only available as a consequence of substituting into a valid formula from •5.
3. Do not penalise trigonometric ratios which are less than −1 or greater than 1 throughout this question.

Commonly Observed Responses:

Candidate C

$f\left(g\left(p\right)\right)=4\sin\left(\sin^{-1}\left(\frac{1}{6}\right)\right)\cos\left(\sin^{-1}\left(\frac{1}{6}\right)\right)$ •5 ✓

$4\times\frac{1}{6}\times\frac{\sqrt{35}}{6}$ •6 ✓

$\frac{\sqrt{35}}{9}$ •7 ✓

[END OF MARKING INSTRUCTIONS]